

Calculate VIX Index from SPX Option Chain

In this assignment, you will write a function program to calculate VIX index from the enclosed data packet. CBOE white paper on VIX calculation is included for your reference.

The SPX option chain data files contain the following information at the time of quote capture:

1. SPX index level
2. Date and Time stamp
3. Quotes for all the strikes for each maturity of the option chain

The TBillQuotes_20231128.csv contains the Treasury bill market quotes. You should

1. build a risk free interest rate curve based on the mid-price of each T-bill quote.
2. Then use interpolation to find the relevant rate to calculate the forward price needed, using dividend yield of 1.56% for SPX index.

Run your program and print out the VIX index you've obtained.

Good luck!